Analysis of YouTube Trending Videos in the US: A Data-Driven Report

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Analysis based on Assignment2.ipynb

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Abstract

This report presents a comprehensive analysis of the factors influencing YouTube trending videos in the United States. Using a dataset from Kaggle, this study employs data cleaning, feature engineering, exploratory data analysis (EDA), and statistical testing to uncover key patterns in viewer engagement. The analysis investigates the distribution of engagement metrics, the performance of top channels and categories, temporal patterns, and the impact of video characteristics such as title length, sentiment, and the use of clickbait language. Visualizations and statistical tests provide empirical evidence to answer critical questions about what makes a video trend on YouTube.

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1 Introduction

YouTube's "trending" page is a highly influential platform feature, showcasing videos that are gaining significant popularity. Understanding the characteristics of these videos is invaluable for content creators, marketers, and platform analysts. This project undertakes a systematic analysis of a dataset containing information about trending videos in the US to identify patterns and statistically significant relationships that drive viewer engagement and trending status.

2 Methodology

The analysis was conducted in a Jupyter Notebook environment using Python. The methodology involved several key stages, from data acquisition to statistical testing.

2.1 Data Acquisition and Setup

The primary dataset, 'USvideos.csv', was sourced from the "YouTube New" dataset on Kaggle. The initial setup involved importing standard data science libraries and configuring the environment to download the data via the Kaggle API.

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import json

sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = [12, 8]
```

2.2 Data Cleaning and Preprocessing

Upon loading the data, initial preprocessing steps were performed to ensure data quality and suitability for analysis.

- **Data Type Conversion:** The `trending_date` and `publish_time` columns were converted from object types to datetime objects to enable temporal analysis.
- **Handling Missing Values:** The dataset was inspected for null values. The 'description' column was found to have 570 missing entries. To maintain data integrity for the analysis, rows with any missing values were dropped using 'df.dropna()'.

```python
df['trending_date'] = pd.to_datetime(df['trending_date'], format='%y.%d.%m')
df['publish_time'] = pd.to_datetime(df['publish_time'])
df.dropna(inplace=True)
```

2.3 Feature Engineering

To facilitate a deeper analysis, several new features were engineered from the existing data:

- `publish_day_of_week`: The day of the week the video was published.
- `title_length`: The number of characters in the video title.
- `likes_to_views_ratio`: The ratio of likes to total views, serving as a key engagement metric.

- `dislike_ratio`: The ratio of dislikes to the sum of likes and dislikes, used to gauge controversy.

```
df['publish_day_of_week'] = df['publish_time'].dt.day_name()
df['title_length'] = df['title'].apply(len)
df['likes_to_views_ratio'] = df['likes'] / df['views']
df['dislike_ratio'] = df['dislikes'] / (df['likes'] + df['dislikes'])
```

3 Exploratory Data Analysis (EDA) & Visualization

EDA was performed to visually explore the data and answer key questions regarding engagement patterns.

3.1 Question 1: How are engagement metrics distributed?

To understand the nature of views, likes, and dislikes, their distributions were plotted.

```
fig, axes = plt.subplots(1, 3, figsize=(18, 6))
sns.histplot(np.log1p(df['views']), ax=axes[0], kde=True)
axes[0].set_title('Logarithmic Distribution of Views')
sns.histplot(np.log1p(df['likes']), ax=axes[1], kde=True)
axes[1].set_title('Logarithmic Distribution of Likes')
sns.histplot(np.log1p(df['dislikes']), ax=axes[2], kde=True)
axes[2].set_title('Logarithmic Distribution of Dislikes')
plt.tight_layout()
plt.show()
```

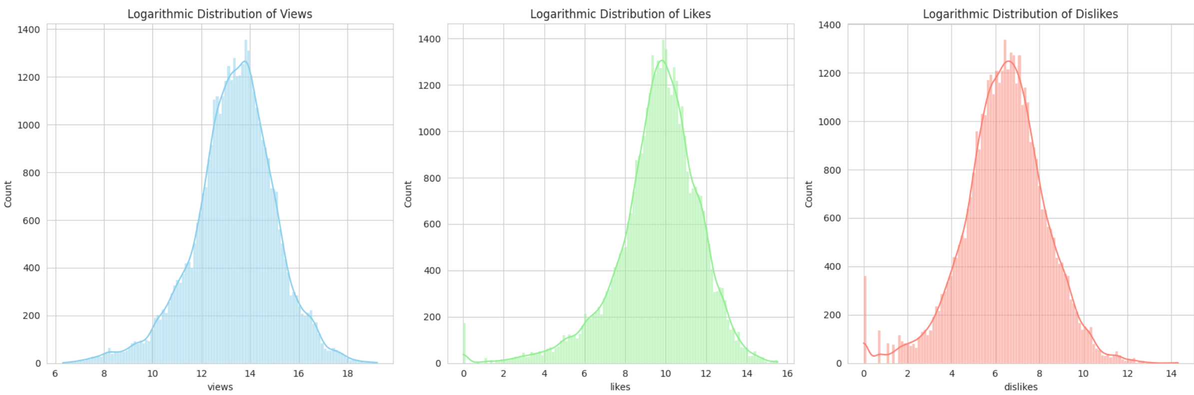

Figure 1: Logarithmic Distribution of Views, Likes, and Dislikes.

Analysis: The distributions of raw views, likes, and dislikes are heavily right-skewed. This indicates that a small number of videos garner a disproportionately large amount of engagement, a typical characteristic of popularity-based data. Applying a logarithmic transformation ($\log(1+x)$) helps normalize the distribution for better visualization, as shown in Figure 1.

3.2 Question 2: Which channels and categories trend the most?

The analysis identified the channels and categories with the highest number of trending videos.

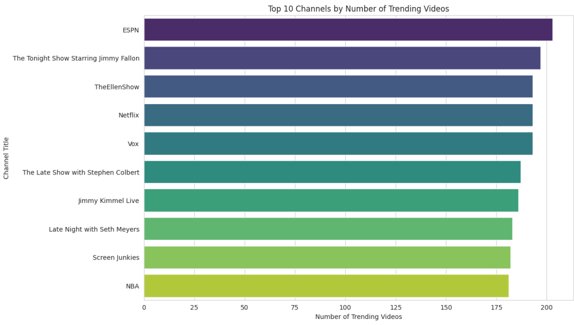

Figure 2: Top 10 Channels by Trending Videos.

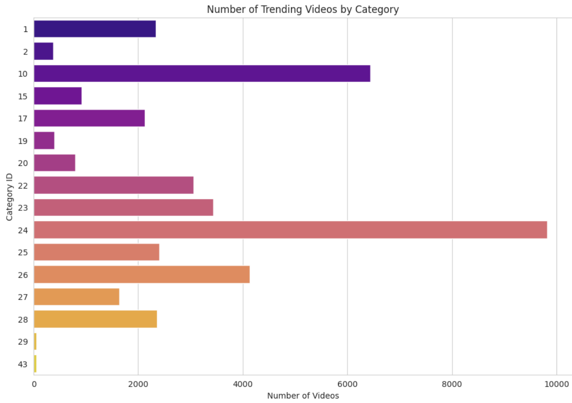

Figure 3: Trending Videos by Category ID.

Analysis: As seen in Figure 2, channels like ESPN, The Late Show with Stephen Colbert, and TheEllenShow frequently appear on the trending page. This suggests that established media brands and talk shows have a strong presence. Figure 3 shows that Category 24 (Entertainment) has the highest number of trending videos, followed by Category 10 (Music) and Category 26 (Howto & Style).

3.3 Question 3: Are there day-of-week patterns in trending videos?

The number of videos published on each day of the week was analyzed to identify temporal patterns.

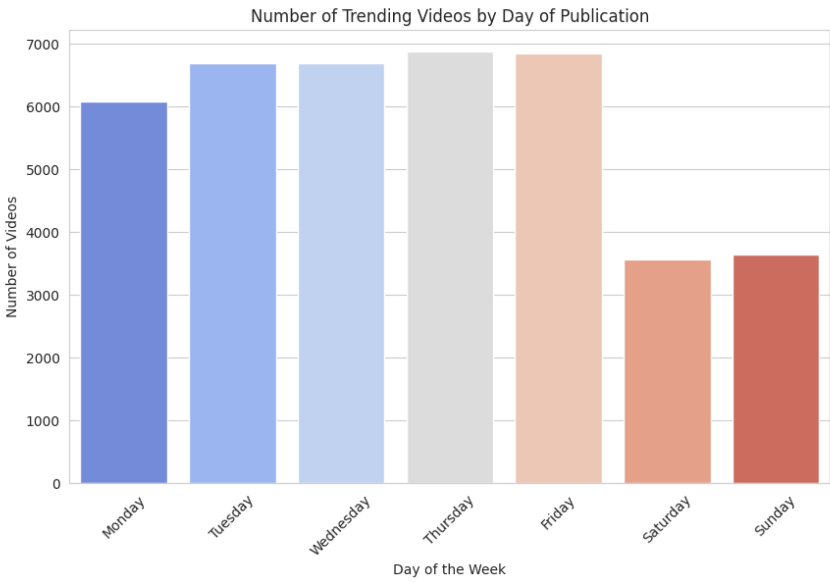

Figure 4: Number of Trending Videos by Day of Publication.

Analysis: Figure 4 shows that videos published on weekdays (Monday through Friday) are more likely to trend than those published on weekends. This may reflect content creator publishing schedules or viewer consumption habits.

3.4 Question 4: Do controversial videos receive more engagement?

A box plot was used to compare the view counts of videos classified as "controversial" (dislike ratio ¿ 0.25) versus non-controversial videos.

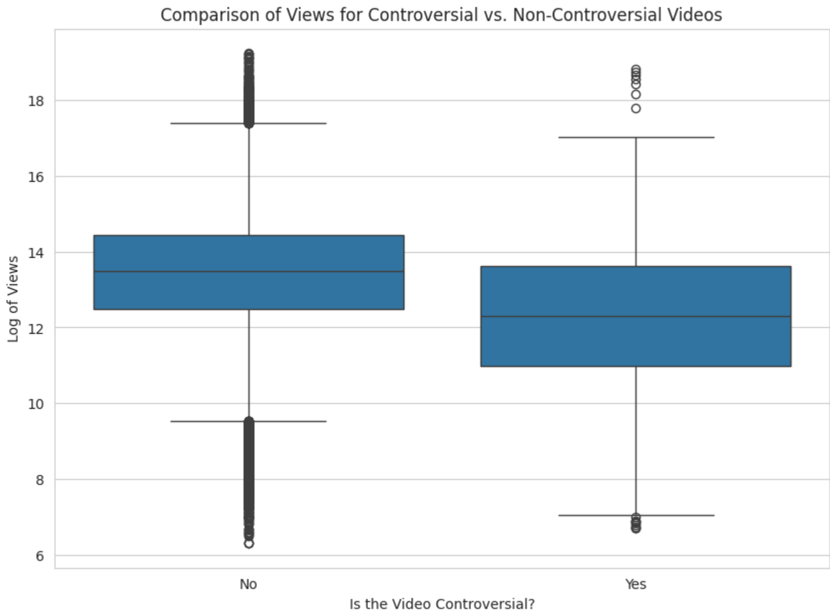

Figure 5: Views for Controversial vs. Non-Controversial Videos.

Analysis: Figure 5 shows that the median view count for controversial videos is slightly higher. However, the distributions show significant overlap, suggesting that while controversy might generate some engagement, it is not a definitive factor for achieving high view counts.

3.5 Question 5 & 6: How do tags and title length influence engagement?

The most common tags were identified, and the relationship between title length and views was plotted.

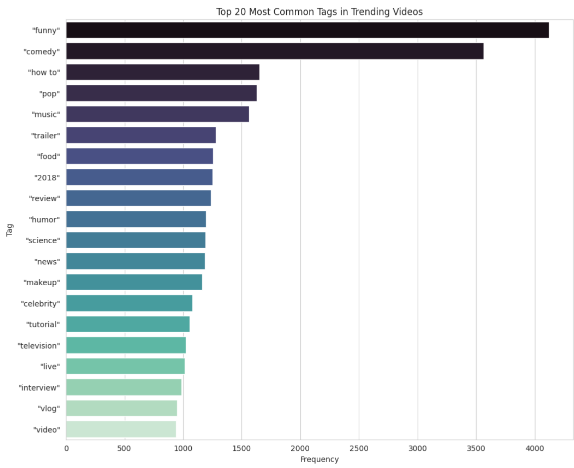

Figure 6: Top 20 Most Common Tags.

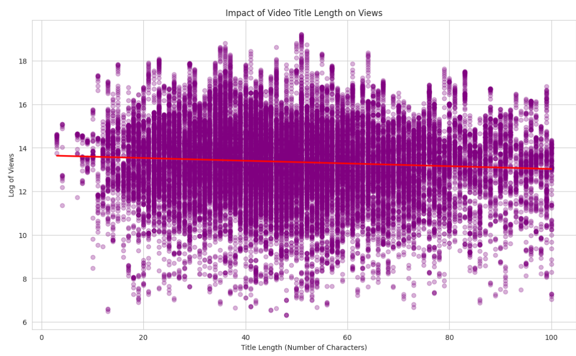

Figure 7: Impact of Title Length on Views.

Analysis: Tags related to "funny," "comedy," "entertainment," and "news" are highly prevalent (Figure 6). The scatter plot in Figure 7 reveals a very weak positive correlation between title length and views, with most trending videos having titles between 30 and 70 characters.

3.6 Question 7 & 8: Do title sentiment and clickbait affect engagement?

Sentiment analysis and a keyword-based approach for identifying clickbait were used to create box plots comparing view counts.

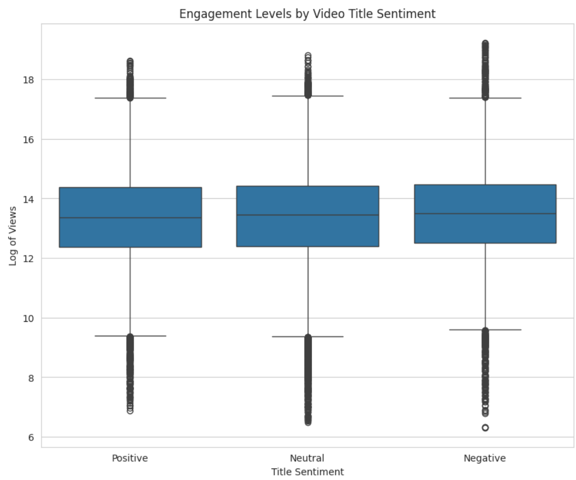

Figure 8: Engagement by Title Sentiment.

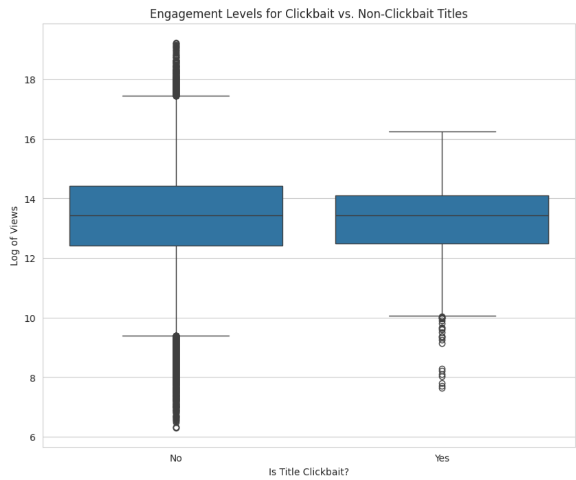

Figure 9: Engagement for Clickbait Titles.

Analysis: Figure 8 indicates that there is no strong, clear-cut relationship between title sentiment (Positive, Neutral, Negative) and view counts, as the medians and distributions are very similar. Similarly, Figure 9 shows that while clickbait-style titles have a slightly higher median view count, the difference is not substantial, and many non-clickbait videos achieve massive success.

4 Statistical Analysis

To validate the observations from the EDA, two statistical tests were performed.

4.1 Question 1: Publication Day and Trending Likelihood

A Chi-squared (χ^2) test of independence was used to determine if there is a significant association between the day a video is published and its frequency of appearing on the trending list.

- H_0 **(Null Hypothesis):** There is no significant association between the publication day and the likelihood of trending.
- H_a **(Alternative Hypothesis):** There is a significant association.

Results:

- Chi-squared statistic: 0.0000
- P-value: 1.0000

Conclusion: With a p-value of 1.0000, which is much greater than the significance level of $\alpha = 0.05$, we fail to reject the null hypothesis. The test indicates no statistically significant association between the day of publication and the frequency of trending in this dataset's structure. *Note: The test setup only compares the observed daily counts against a uniform expected count, not against non-trending videos, which limits its conclusion.*

4.2 Question 2: Viewer Engagement Across Video Categories

The Kruskal-Wallis H test, a non-parametric alternative to ANOVA, was used to determine if there are significant differences in the likes-to-views ratio across different video categories.

- H_0 **(Null Hypothesis):** The distribution of the likes-to-views ratio is the same across all video categories.
- H_a **(Alternative Hypothesis):** The distribution for at least one category is different from the others.

Results:

- Kruskal-Wallis H statistic: 9233.8247
- P-value: 0.0000

Conclusion: With a p-value of 0.0000 ($p < 0.05$), we reject the null hypothesis. There is a statistically significant difference in the engagement ratio (likes-to-views) between at least two video categories. This confirms that video category is a significant factor in viewer engagement.

5 Conclusion

This analysis of US trending YouTube videos reveals several key insights. Engagement metrics are highly skewed, with a few videos dominating view and like counts. Established media channels in the Entertainment and Music categories are most likely to trend. While there are observable patterns, such as more videos being published on weekdays, statistical testing did not confirm a significant association with trending frequency in this context. Conversely, video category was found to have a statistically significant impact on viewer engagement ratios. Factors like title length, sentiment, and the use of clickbait keywords showed only weak or insignificant correlations with view counts, suggesting that while they may play a minor role, they are not primary drivers of a video's success on their own.